IQRA University (IU)		
Faculty of Engineering Sciences and Technology (FEST)		
Department of Computer Science		
Course Code	**Course Name**	**Credit Hr**
CSC412	**Information Security**	**3+1**

1. Basic Information			
Instructor	Dr. Sorath Mahar	**Designation**	Assistant Professor
Prerequisite(s)	Computer Networks	**Semester**	Summer 2026
Email	Sorath.mahar@iqra.edu.pk	**Phone**	--
Office Hours	11:30 – 01:00 (Mon- Fri)	**Office Location**	Faculty Room 5A – 5th Floor

2. Course Objective(s)

This course provides a broad overview of the threats to the security of information systems, the responsibilities and basic tools for information security, and the levels of training and expertise needed in organizations to reach and maintain a state of acceptable security. It covers concepts and applications of system and data security. Areas of particular focus include secure network design, implementation and transition issues, and techniques for responding to security breaches.

3. Course Contents

Information security foundations, security design principles; security mechanisms, symmetric and asymmetric cryptography, encryption, hash functions, digital signatures, key management, authentication and access control; software security, vulnerabilities and protections, malware, database security; network security, firewalls, intrusion detection; security policies, policy formation and enforcement, risk assessment, cybercrime, law and ethics in information security, privacy and anonymity of data.

4. Course Learning Outcomes

CLOs	CLO Statement	BT Level	Mapping			% Weight
			GAs	ACM KA	SGDs	
CLO1	**Explain** the key concepts of Information Security that includes design principles, cryptography, risk management and ethics	C2	GA2	#7 IAS	4	-
CLO2	**Analyze** appropriate techniques to tackle and solve problems in the discipline of information security	C4	GA3			-
CLO3	**Design** security strategies using various tools and devices for achieving information security and privacy such as Network Firewall and VPN's using IP Sec and SSL protocols	C6	GA4			-

Note: On successful completion of course GA 1 (Academic Education) will automatically attain.

5. Course Textbook / Reference Books and Supplementary Reading Material

S No	Book Title	Author(s)	Edition/ publication year/publisher
1.	Computer Security: Principles and Practice	William Stallings	4th edition, 2019, Pearson
2.	Principles of Information Security	M. Whitman and H. Mattord	6th edition. 2018, Course Technology
3.	Computer Security	Dieter Gollmann	3rd edition, 2011, John Wiley & Sons

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6. CLO Outcome Based Assessment (OBA)						
Assessment Tool		**CLO Mapped**	**CLO Marks**	**% Weight**	**Total Marks**	**Assessment Date**
Quizzes	Quiz #1	*CLO1*	3	30%	3	
	Quiz #2	*CLO2*	3	30%	3	
10	Quiz #3	*CLO3*	4	40%	4	
	Total Quizzes %			**100%**	10	
Assignments	Assignment #1	*CLO1*	5	33.3%	5	
	Assignment #2	*CLO2*	5	33.3%	5	
15	Assignment #3	*CLO3*	5	33.3%	5	
				100%	15	
Midterm	Midterm Q#1	*CLO1*	6	24%	**6**	
25	Midterm Q#2	*CLO2*	7	28%	**7**	
	Midterm Q#3	*CLO2*	6	24%	**6**	
	Midterm Q#4	*CLO3*	6	24%	**6**	
	Total Midterm %			**100%**	**25**	
Project/CCP 10	Project/CCP	*CLO3*	10			
	Total Project /CCP %			**100%**	**10**	
Final Exam	Final Exam Q#	*CLO1/2/3*	10			
	Final Exam Q#	*CLO1/2/3*	10			
40	….	*CLO1/2/3*				
	Final Exam Qn	*CLO1/2/3*				
	Total Final Exam %			**100%**		
100				**Total Marls**	**100**	
Note: Please make sure every CLO must be assessed at least 3 time.						

7. Weekly Plan				
Week	**Lecture No**	**Topic Covered**	**CLO**	**Assessment Tool**
1	1	Introduction to Information Security, Key Factors	**CLO/1/2/3**	**Quiz#, Mid, Final**
	2	Types and Classes of attacks, Interruption, Interception, Modification, Fabrication	1	
	3	Active vs Passive Attacks, Brute force Attack, Numerical	1	
2	4	Encryption, Types of Operation, Substitution/ Transposition cipher, Block vs Stream ciphers	1	
	5	Ceasar Cipher, Mono Alphabetic Cipher, Playfair Cipher	1	
	6	Rail Fence Cipher, Data Encryption Standard (DES)	1	
3	7	Data Encryption Standard (DES)	1	
	8	Advanced Encryption Standard (AES)	1	
	9	Symmetric Key Algorithm and its Drawbacks	2	
4	10	Public Key Cryptography, Asymmetric Key Algorithm	2	
	11	Applications of Public Key Algorithms	2	
	12	Mathematics in Cryptography, Trapdoor Function	2	
5	13	RSA Algorithm, Numerical	2	
	14	Diffie Hellman Key Exchange	2	
	15	Data Integrity, MD5 and SHA Hash Algorithm	2	
6	16	Application of Hash Algorithm (Password Storage)	2	

	17	Digital Signature, Email Security Pretty Good Privacy (PGP)	2	
	18	Web Security, Secure Socket Layer (SSL)	2	
7	19	How SSL/TLS works, Certificate Signing Request (CSR), Self-Signed Certificate	3	
	20	Project/CCP	3	
	21	Project/CCP	3	
8	**Midterm Exam**			
9	22	Private/Public IP address, NAT/PAT	3	
	23	Firewall Basics, Security zones	3	
	24	Stateless vs Stateful Firewall, Basic rules of the Firewall	3	
10	25	Cisco ASA Firewall configuration, Configuring NAT/PAT	3	
	26	Access Control Lists (ACL)	3	
	27	DDoS/SYN Attack and Prevention	3	
11	28	Application Layer Inspection	3	
	29	IP Security	3	
	30	Virtual Private Networks – VPN , Encapsulation	3	
12	31	Types and Application of VPN	2	
	32	Authentication methods, AAA Authentication	2	
	33	RADIUS Server	2	
13	34	Intrusion Prevention and Detection System IPS/IDS, Classes of Intruders	3	
	35	Network vs Host based IPDS, Detection methods	3	
	36	Signature vs Anomaly based IPDS and their drawbacks	3	
14	37	Availability of Data (Fault Tolerance by implementing RAID)	2	
	38	Disaster Recovery and Backup Strategies	2	
	39	Malicious Software, Spam, Virus	2	
15	40	Backdoor, Hoaxes, Password Crack, Trojan, Zombie	2	
	41	Threat, Attack, Vulnerability, Risk, Spoofing, Man in the Middle	2	
	42	Risk Assessment, Cyber crime	2	
16	43	Law and Ethics in Information Security	2	
	44	Project/CCP Assessment	3	
	45	Project/CCP Assessment	3	
17	**Final Exam**			

8. IU Assessment / grading Policy		**Instructor grading for course ***
Quizzes	10-15%	10%
Assignments	10-15%	15%
Projects/Presentation/CCP	0-10%	10%
Mid Semester Examination/	20-30%	25%
End Semester Examination	40-50%	40%